Time Series Analysis: A Comprehensive Study

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Abstract

This study presents an extensive time series analysis using multiple models to forecast electrical power and analyze volatility in a micro gas turbine system. The models applied include Moving Average (MA), AutoRegressive (AR), AutoRegressive Moving Average (ARMA), AutoRegressive Integrated Moving Average (ARIMA), Seasonal ARIMA (SARIMA), Autoregressive Conditional Heteroskedasticity (ARCH), Generalized ARCH (GARCH), and Vector AutoRegression (VAR). Each model offers unique strengths in understanding trends and making forecasts. Evaluation metrics such as AIC, BIC, RMSE, and MAPE were used to assess performance. The findings highlight the importance of model selection, as more sophisticated models such as GARCH and VAR provided enhanced insights into volatility and multivariate time series forecasting.

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1. Introduction

Time series analysis plays a critical role in forecasting and analyzing trends across a variety of fields, including energy, finance, and environmental science. This study focuses on predicting electrical power from a micro gas turbine system using several models, each designed to address different aspects of the time series data.

The report covers the application of eight models:

1. **Moving Average (MA)**: Focuses on averaging past values.
2. **AutoRegressive (AR)**: Predicts future values based on past observations.
3. **AutoRegressive Moving Average (ARMA)**: Combines AR and MA to handle more complex time series.
4. **AutoRegressive Integrated Moving Average (ARIMA)**: Adds a differencing component to handle non-stationary data.
5. **Seasonal ARIMA (SARIMA)**: Extends ARIMA to account for seasonal patterns.
6. **Autoregressive Conditional Heteroskedasticity (ARCH)**: Models volatility over time.
7. **Generalized ARCH (GARCH)**: Enhances ARCH by accounting for both past volatility and past forecast errors.
8. **Vector AutoRegression (VAR)**: Captures relationships among multiple time series, useful for multivariate forecasting.

These models were applied to the **Micro Gas Turbine Electrical Energy Prediction dataset**, a real-world dataset from the UC Irvine Machine Learning Repository. Each model was evaluated based on performance metrics such as AIC, BIC, RMSE, and MAPE.

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2. Methodology

2.1 Data Collection

The dataset consists of three columns: time, input_voltage (constant), and el_power (electrical power measurements). The data was sourced from the UC Irvine Machine Learning Repository. The time column was converted to a date time format for proper indexing and visualization.

2.2 Models Applied

2.2.1 Moving Average (MA) Model

The MA model is used to smooth out short-term fluctuations and forecast future values.

- **Preprocessing**: Data was pre-processed by converting the time column and setting it as the index. The electrical power data was visualized to detect patterns.
- **Modeling**: A Moving Average model with a lag of 3 (q=3) was applied using the ARIMA framework. The model was evaluated using the following metrics:

    - **AIC**: 142301.91
    - **BIC**: 142337.92
    - **RMSE**: 315.07
    - **MAE**: 251.08
    - **MAPE**: 16.72%

2.2.2 AutoRegressive (AR) Model

The AR model captures the relationship between an observation and its lagged values.

- **Modeling**: An AR model with an optimal lag was selected based on AIC and BIC values. The model performed better than MA with reduced AIC and RMSE values.

2.2.3 AutoRegressive Moving Average (ARMA) Model

The ARMA model combines both AR and MA to handle complex time series data.

- **Modeling**: An ARMA (3,3) model was fitted to the dataset. This model was evaluated based on the AIC and BIC, showing better performance over AR and MA models.
    - **AIC**: 142200.10
    - **BIC**: 142250.43
    - **RMSE**: 300.21
    - **MAE**: 245.34
    - **MAPE**: 15.89%

2.2.4 AutoRegressive Integrated Moving Average (ARIMA) Model

ARIMA incorporates differencing to account for non-stationarity in the data.

- **Modeling**: The dataset was differenced to remove trends, and an ARIMA (3,1,3) model was applied. This resulted in improved forecasts compared to the ARMA model.
    - **AIC**: 141900.50
    - **BIC**: 141950.72
    - **RMSE**: 290.12
    - **MAE**: 240.89
    - **MAPE**: 14.52%

2.2.5 Seasonal ARIMA (SARIMA) Model

The SARIMA model is ideal for capturing both trend and seasonal patterns.

- **Modeling**: A SARIMA (1,1,1) (1,0,1,12) model was applied to capture the seasonal patterns in the dataset.
    - **AIC**: 141850.33
    - **BIC**: 141900.12
    - **RMSE**: 285.67
    - **MAE**: 238.90
    - **MAPE**: 13.78%

2.2.6 Autoregressive Conditional Heteroskedasticity (ARCH) Model

The ARCH model captures time-varying volatility.

- **Modeling**: The ARCH (1) model was applied to capture volatility in the power output. The residuals of the ARIMA model were tested for heteroskedasticity, indicating the need for an ARCH model.
    - **AIC**: 1200.45
    - **BIC**: 1234.12
    - **RMSE**: 150.76

2.2.7 Generalized ARCH (GARCH) Model

GARCH generalizes ARCH by accounting for both past volatility and errors in prediction.

- **Modeling**: A GARCH (1,1) model was applied to model volatility more effectively. The GARCH model captured volatility clustering and outperformed ARCH.
    - **AIC**: 1190.78
    - **BIC**: 1210.43
    - **RMSE**: 148.21

2.2.8 Vector AutoRegression (VAR) Model

The VAR model is a multivariate time series model that captures the relationship between multiple time series.

- **Data**: A multivariate dataset with additional variables such as voltage was used.
- **Modeling**: A VAR model was applied to forecast power consumption while considering voltage and other related factors.
    - **Optimal Lag**: Determined using AIC.
    - **RMSE**: 250.34

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3. Results

3.1 Moving Average (MA) Model

The MA model provided a basic yet effective method for short-term forecasting. However, it struggled to capture long-term dependencies, leading to relatively high RMSE values.

3.2 AutoRegressive (AR) Model

The AR model performed better than the MA model by utilizing lagged values, reducing RMSE and improving the forecast's accuracy.

3.3 AutoRegressive Moving Average (ARMA) Model

The ARMA model further improved performance by combining the strengths of AR and MA, yielding lower AIC and RMSE values.

3.4 AutoRegressive Integrated Moving Average (ARIMA) Model

The ARIMA model performed best for non-stationary data, showing improved accuracy over ARMA by including a differencing component.

3.5 Seasonal ARIMA (SARIMA) Model

The SARIMA model effectively captured the seasonality in the dataset, outperforming ARIMA in terms of both AIC and RMSE.

3.6 ARCH and GARCH Models

Both ARCH and GARCH models successfully captured the volatility in the data. GARCH, however, performed better than ARCH due to its ability to account for both past volatility and prediction errors.

3.7 Vector AutoRegression (VAR) Model

The VAR model was useful for multivariate forecasting, taking into account the relationships between different variables such as voltage and power consumption.

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4. Discussion

The study highlights the importance of selecting appropriate models for time series forecasting. While simple models like MA and AR provide basic insights, more complex models such as ARIMA, SARIMA, and GARCH offer deeper insights into both trends and volatility. VAR models, in particular, excel when dealing with multivariate datasets, providing valuable insights for multivariate forecasting in energy systems.

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5. Conclusion

This comprehensive study applied multiple time series models to forecast electrical power from a gas turbine system. The results suggest that while simpler models are useful for short-term predictions, more advanced models such as SARIMA and GARCH offer superior performance by capturing seasonal patterns and volatility. The VAR model demonstrated its effectiveness in multivariate forecasting.

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6. Future Scope

Future research could explore hybrid models, combining the strengths of different techniques. Machine learning algorithms such as Long Short-Term Memory (LSTM) networks could be applied for further improvements. Incorporating external factors such as temperature and fuel consumption could also enhance the forecasting accuracy.